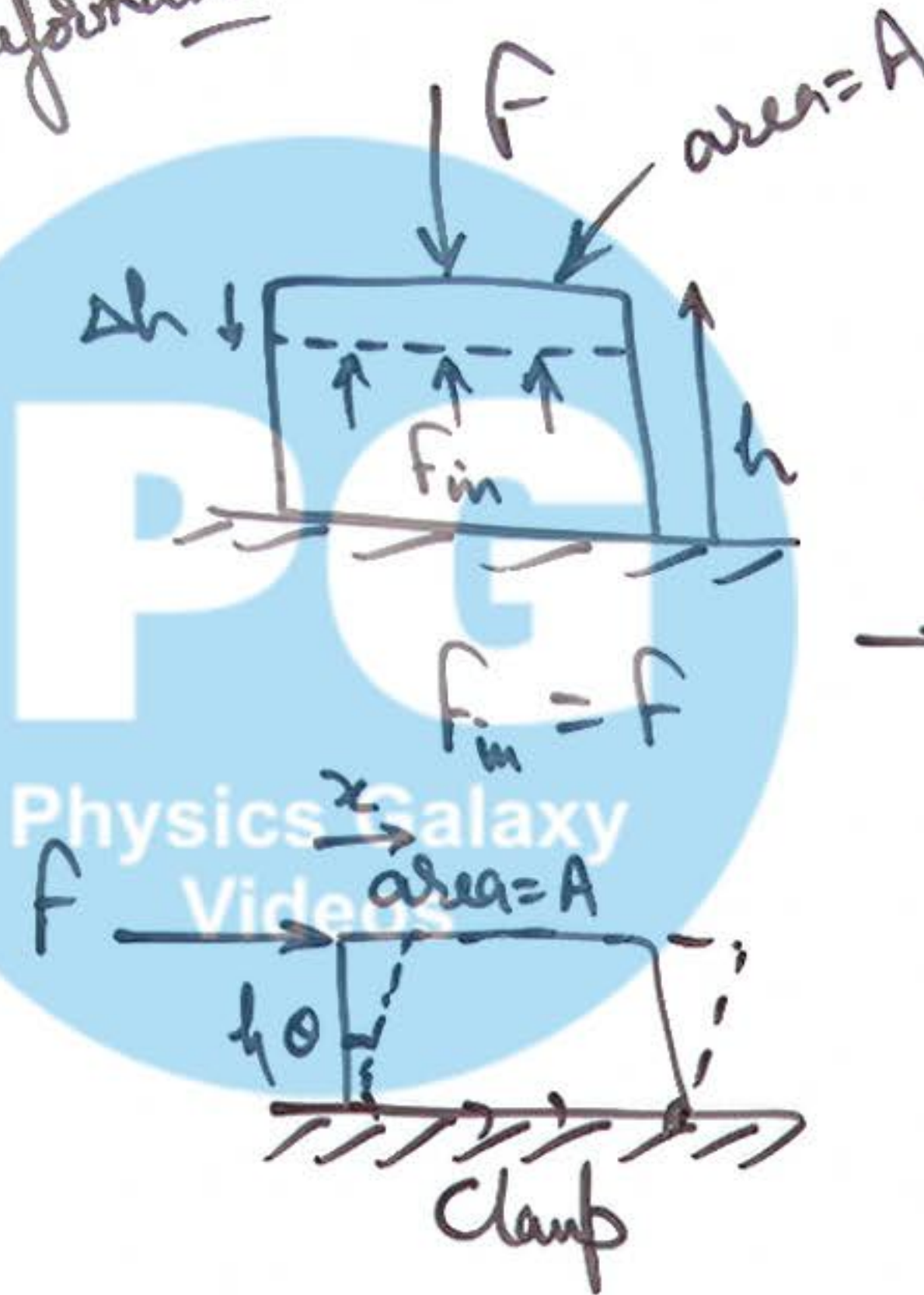


① Stress & Strain in elastic material:

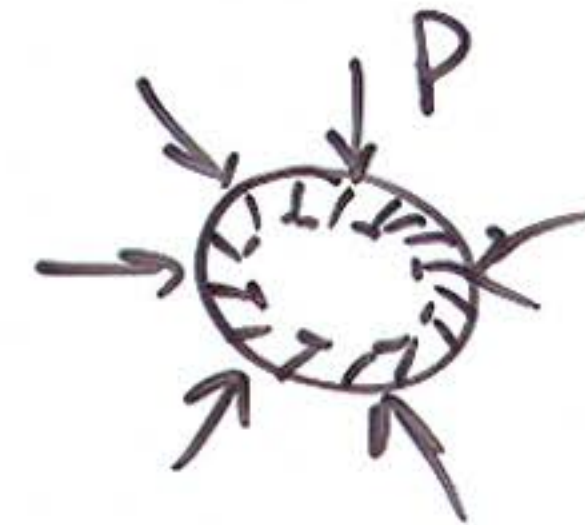
→ internal force / Area.

relative deformation



$$\text{Normal Strain} = \frac{\Delta h}{h}$$

$$\text{Normal Stress} = \frac{F}{A}$$

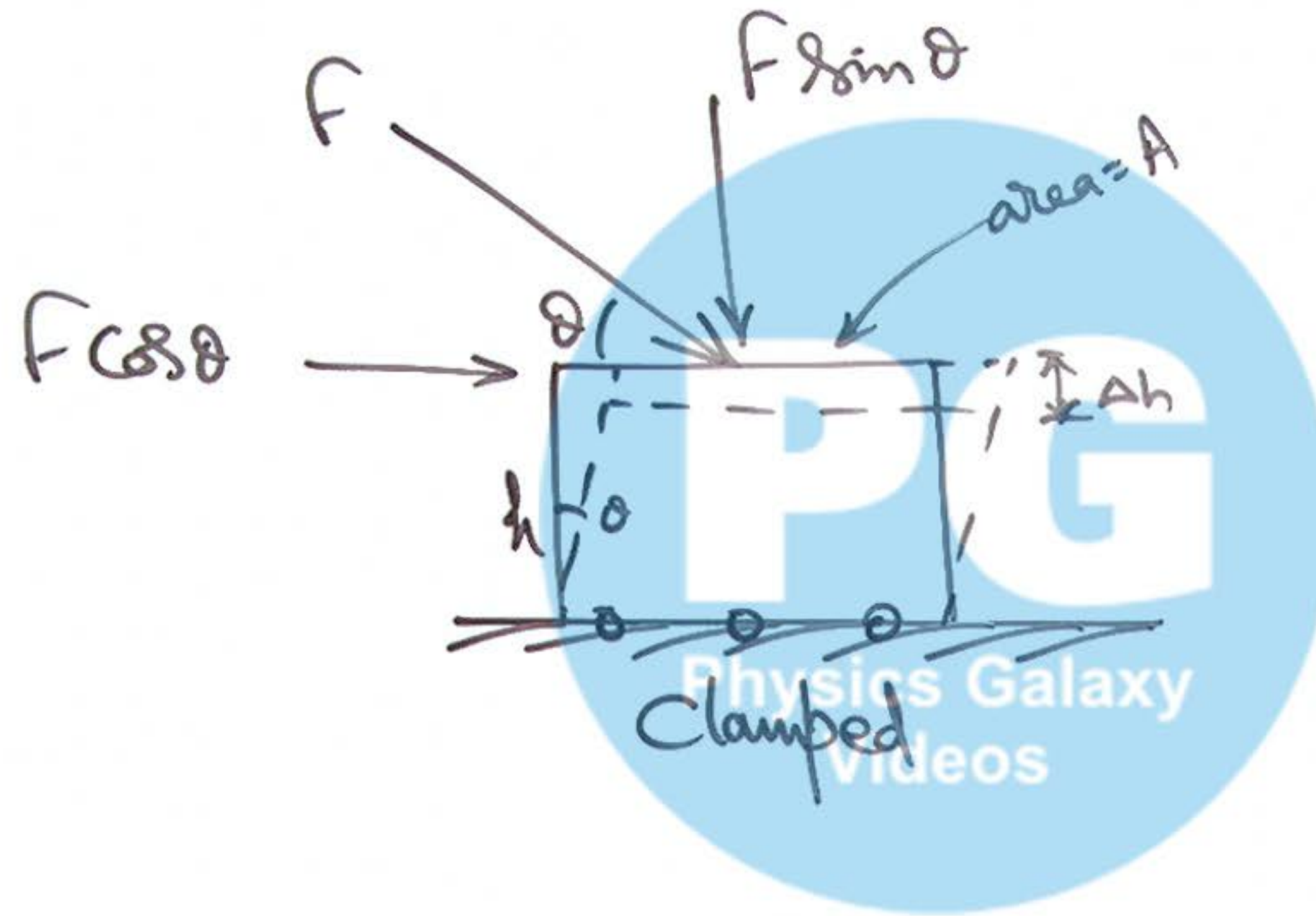


$$\text{Volume Stress} = P_{ex.}$$

$$\text{Volume Strain} = \frac{\Delta V}{V}$$

$$\text{Shear Stress} = \frac{F}{A}$$

$$\text{Shear Strain } \theta = \frac{x}{h}$$



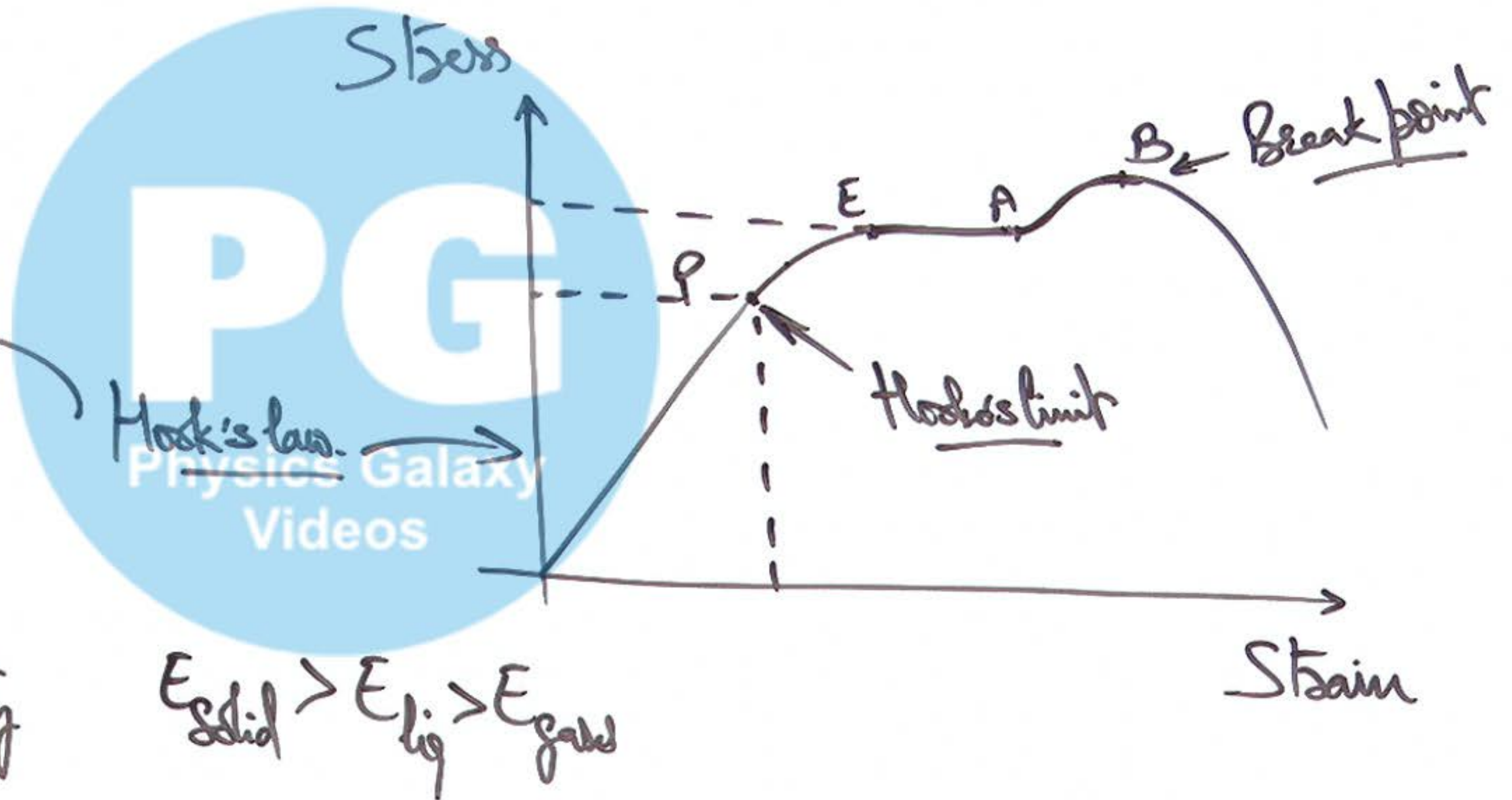
longitudinal/normal
 stress = $\frac{F \sin \theta}{A}$

Shear stress = $\frac{F \cos \theta}{A}$

② Relation betwⁿ Stress & Strain:

$$E = \frac{\text{Stress}}{\text{Strain}}$$

↓
Constant of elasticity
& modulus of elasticity



③ Types of Elasticity Modulus:



① Young's Modulus for solids

$$Y = \frac{\text{Longitudinal Stress}}{\text{Long. Strain}} = \frac{F/A}{\Delta l/l}$$

Compressibility

$$C = \frac{1}{B}$$

② Bulk Modulus for liq

$$B = \frac{\text{Volume Stress}}{\text{Volume Strain}} = \frac{\Delta P_{\text{ex}}}{\Delta V/V}$$

③ Shear Modulus

$$\eta = \frac{\text{Shear Stress}}{\text{Shear Strain}} = \frac{F/A}{\theta}$$

$$Y = \frac{Mg/A}{\Delta l/l} \rightarrow \boxed{\Delta l = \frac{Mg \cdot l}{YA}}$$

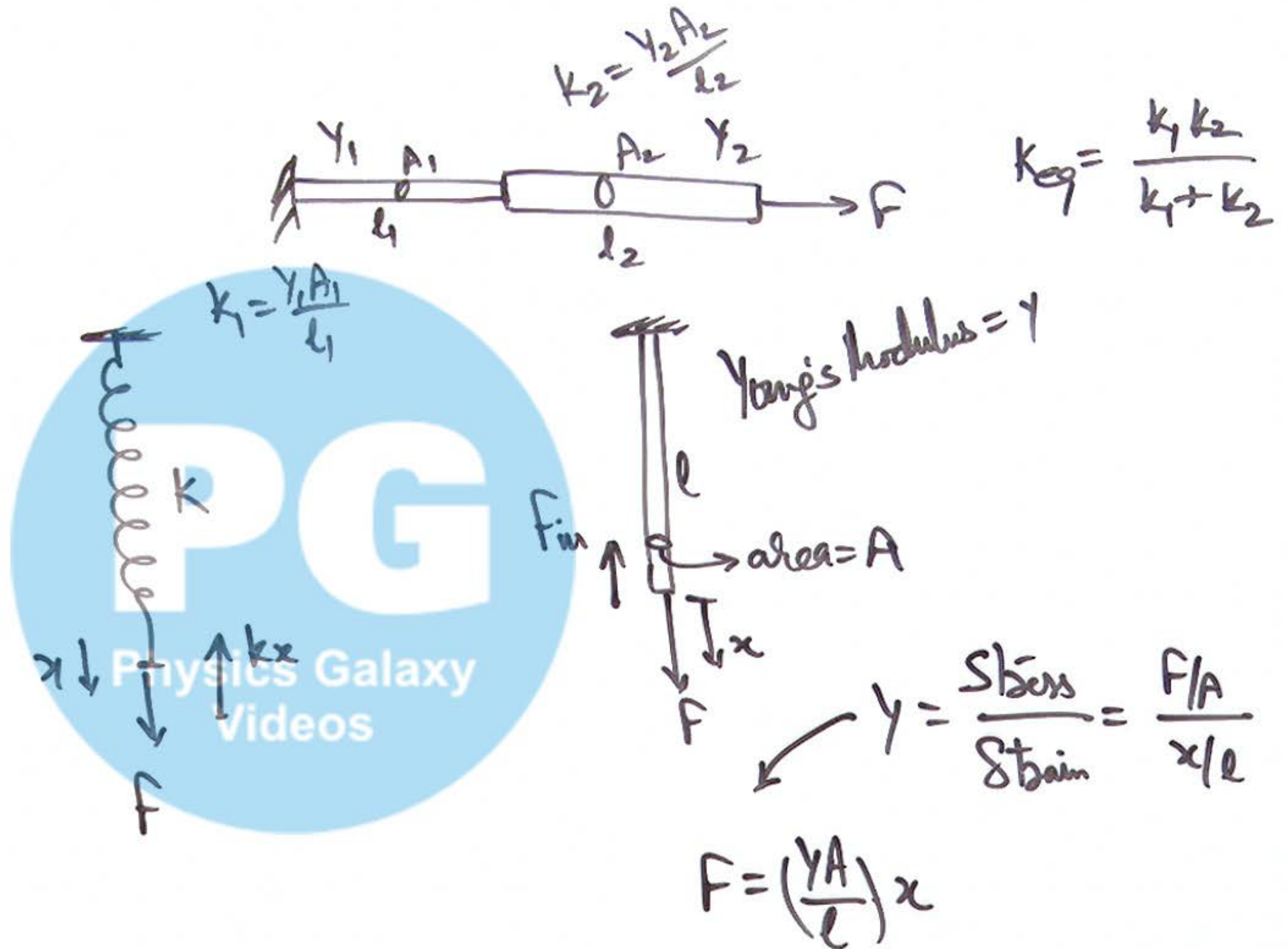
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④ Metal Wires or Rods as Equivalent Springs:

$$F = kx$$

$$\downarrow$$

$$\left(\frac{YA}{l}\right)$$



⑤ Potential Energy due to Elasticity:

PE density in
deformed elastic material

$$u_e = \frac{1}{2} \times \text{Stress} \times \text{Strain} \quad \text{J/m}^3$$

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